

Testing – NCEA Specifications

NZQA requirement :

Excellence – Comprehensive testing

What is testing?

- Process of examining if your digital outcome works correctly and meets requirements.
- Purpose:
 - To prove your outcome is fit for purpose

Requirements are met

E.g. Site must have a menu

Specifications are followed

E.g. The password field must require at least 8 characters and one special symbol

Basic functionality is solid

E.g. links work, code doesn't crash, buttons do what they should

- Iterative process – sprint testing

Testing categories and types

There are two categories of testing based on the input data

- Program testing
- SQL/Database testing

Program testing types

			Excellence	Merit	Achieved
Testing Requirement	Focus				
Expected/normal testing	Showing the outcome works as expected under normal conditions	Example: Instagram – Caption field - A text box where a user types a description for their photo (20 characters) Test - Typing "My summer holiday!" and clicking post. Result - The text appears exactly as typed when you hit 'Post'			
Boundary testing	Testing values at the exact limits	Test – leave the caption empty (lower boundary) or paste exactly 20 characters Result – both boundaries should work and the text must be shown			
Invalid testing	Testing data that should fail	Test – Type more than 20 characters Result – Your program must show an error message “Caption too long”			

SQL testing types

Testing Requirement	Focus	Example: Instagram
Relational Testing	Testing that data correctly links across multiple tables	Test: When a user deletes their account, verify that all their Photos and Comments are also removed from the database (Cascading Delete)
Data Modification	Verifying that INSERT, UPDATE, and DELETE queries correctly modify the data without errors	Test: Change a post caption from "Old Caption" to "New Caption". Verify the UPDATE query changes the text in the database without affecting the image URL
Data Validation	Checking that form controls (like dropdowns or text boxes) enforce validation rules correctly	Test: Attempt to upload a photo with a caption exceeding 2,200 characters. Verify the UI blocks the post or the database rejects the INSERT due to character limits

Evidence of testing – NCEA requirements

- Good evidence includes:
 - Annotated screenshots – before and after testing
 - Testing tables – showing what was tested, expected result, actual result, steps taken
 - Screenshots/videos – short clips showing program running and handling different inputs
 - Debugging examples – Descriptions or code snippets showing how you identified and resolved errors

Testing vs Trialling

	Testing	Trialing
Who	Developer	End-Users/other
Purpose	To ensure functionality, data integrity, and that it works as intended	To test usability, suitability, and user experience (UX)
Focus	"Does it work?" (e.g., clicks, links, code)	"Is it fit for purpose?" (user feedback)
Example	Checking that a form only accepts numbers	Asking a classmate to find information on a site
NCEA Level	Essential for Achieved	Required for Merit/Excellence.

Testing table – Instagram context

Test Type	Test Case	Input/Action	Expected result	Actual result	Pass/ Fail	Improvement
Category: Program Type: Invalid	Caption Character Limit (20)	Type a caption with 23 characters	The "Share" button becomes disabled and a red "Limit Exceeded" warning appears	Button stayed active; user could still click share.	Fail	Added an if statement to check caption.length and disable the button
Category: SQL Type: Relational	Deleting a User Account	Run: DELETE FROM Users WHERE UserID = '101'	The user is deleted AND all their photos in the Posts table are automatically removed	User was deleted, but their photos remained in the database	Fail	Updated the Database Schema to include ON DELETE CASCADE on the Foreign Key
Category: SQL Type: Data Validation	Username Length	Create a username with exactly 30 characters (the max limit)	Database accepts the name and saves it successfully	Error: "Data too long for column 'username'"	Fail	Changed SQL data type from VARCHAR(20) to VAR CHAR(30) to match requirements